# Abstract Algebra by Pinter, Chapter 23, question B3

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Chapter 23 on Number Theory

## Contents

**Proof** **1**
Initial Question . . . 1
Simultaneous Solution for Three Elements . . . 1
Generalizing to $k+1$ through induction . . . 3
Relation between gcd and lcm operators . . . 3
Proving equivalency holds under gcd for $k+1$ . . . 3
Generalizing lcm to Multiple Arguments . . . 3
Solution $c \equiv c_i$ is also a Solution in the lcm of the gcds . . . 4
There is a Common Solution for $c$ and $c_{k+1}$ . . . 4

## Proof

### Initial Question

*We are given k congruences*

$$x \equiv c_1 (\mathrm{mod}\, m_1) \qquad x \equiv c_2 (\mathrm{mod}\, m_2) \qquad \cdots \qquad x \equiv (\mathrm{mod}\, m_k)$$

*for all* $i, j \in \{1, \ldots, k\}$

$$c_i \equiv c_j (\mathrm{mod}\, d_{ij})$$

*where* $d_{ij} = \gcd(m_i, m_j)$.

*Prove there is an x satisfying all k congruences simultaneously, and the solution is of the form*

$$x \equiv c (\mathrm{mod}\, t)$$

*where* $t = \mathrm{lcm}(m_1, m_2, \ldots, m_k)$.

### Simultaneous Solution for Three Elements

We will proceed to prove these statements through induction, first starting with the case of proving there is a simultaneous solution for $c_1, c_2$ and $c_3$.

It has been shown earlier in theorem 3 that there is a solution for two equations $x \equiv a (\mathrm{mod}\, n)$ and $x \equiv b (\mathrm{mod}\, m)$, only exists if

$$a \equiv b (\mathrm{mod}\, d)$$

$$d = \gcd(m, n)$$

For the first two equations, there is therefore a simultaneous solution because

$$c_1 \equiv c_2 (\mathrm{mod}\, d_{12})$$

Earlier in theorem 4, it was shown that if $x \equiv a (\mathrm{mod}\, n)$ and $x \equiv b (\mathrm{mod}\, m)$ have a simultaneous solution, it is of the form

$$x \equiv c (\mathrm{mod}\, t)$$

$$t = \mathrm{lcm}(m, n)$$

So therefore the solution of $x \equiv c_1 (\mathrm{mod}\, m_1)$ and $x \equiv c_2 (\mathrm{mod}\, m_2)$ is

$$x \equiv c (\mathrm{mod}\, t)$$

$$t = \mathrm{lcm}(m_1, m_2)$$

We want to know if there is a solution $x$ for $x = c (\mathrm{mod}\, t)$ and $x = c_3 (\mathrm{mod}\, m_3)$. That is whether the statement

$$c_3 \equiv c [\mathrm{mod}\, \mathrm{gcd}(t, m_3)]$$

is true.

But we know that $\mathrm{gcd}(a, \mathrm{lcm}(b, c)) = \mathrm{lcm}(\mathrm{gcd}(a, b), \mathrm{gcd}(a, c))$ so

$$\begin{aligned} \mathrm{gcd}(t = \mathrm{lcm}(m_1, m_2), m_3) &= \mathrm{lcm}(\mathrm{gcd}(m_1, m_3), \mathrm{gcd}(m_2, m_3)) \\ &= \mathrm{lcm}(d_{13}, d_{23}) \end{aligned}$$

So we want to know whether this statement is true

$$\begin{aligned} c_3 &\equiv c [\mathrm{mod}\, \mathrm{gcd}(t, m_3)] \\ &\equiv c [\mathrm{lcm}(d_{13}, d_{23})] \end{aligned}$$

At the start it was stated that $c_3 \equiv c_1 (\mathrm{mod}\, d_{13})$, and we also we know that

$$c \equiv c_1 (\mathrm{mod}\, m_1) \implies c \equiv c_1 (\mathrm{mod}\, d_{13})$$

$$\therefore c \equiv c_3 (\mathrm{mod}\, d_{13})$$

Likewise $c_3 \equiv c_2 (\mathrm{mod}\, d_{23}) \implies c \equiv c_3 (\mathrm{mod}\, d_{23})$

Now from the last part of theorem 4, we note that

$$m \mid (x - c) \text{ and } n \mid (x - c) \iff t \mid (x - c)$$

or

$$x \equiv c (\mathrm{mod}\, m) \text{ and } x \equiv c (\mathrm{mod}\, n) \iff x \equiv c (\mathrm{mod}\, t)$$

Note that

$$d_{13} \mid (c - c_3) \text{ and } d_{23} \mid (c - c_3) \iff lcm(d_{13}, d_{23}) \mid (c - c_3)$$

or

$$c \equiv c_3 (\mathrm{mod}\, d_{13}) \text{ and } c \equiv c_3 (\mathrm{mod}\, d_{23}) \iff c \equiv c_3 [\mathrm{mod}\, \mathrm{lcm}(d_{13}, d_{23})]$$

That is we can state that

$$c_3 \equiv c [\mathrm{mod}\, \mathrm{lcm}(d_{13}, d_{23})]$$

But $\mathrm{lcm}(d_{13}, d_{23}) = \mathrm{gcd}(t, m_3)$. So by theorem 3 because

$$c_3 \equiv c [\mathrm{mod}\, \mathrm{gcd}(t, m_3)]$$

there is a simultaneous solution of

$$\begin{aligned} x &\equiv c (\mathrm{mod}\, t) \\ x &\equiv c_3 (\mathrm{mod}\, m_3) \end{aligned}$$

And this is also the solution for

$$\begin{aligned} x &\equiv c_1 (\mathrm{mod}\, m_1) \\ x &\equiv c_2 (\mathrm{mod}\, m_2) \end{aligned}$$

## Generalizing to $k+1$ through induction

Now we will generalize this using induction on $k+1 \in \mathbb{Z}$ terms where we assume $S_k$ is true, proving the statement $S_{k+1}$ is true, and therefore it is true for all integers.

Assume there is a solution of $k$ congruences

$$x \equiv c_1 (\mathrm{mod}\, m_1) \qquad \cdots \qquad x \equiv c_k (\mathrm{mod}\, m_k)$$

of the form

$$x \equiv c (\mathrm{mod}\, t)$$
$$t = \mathrm{lcm}(m_1, \ldots, m_k)$$

Note that $\forall i, j \in \{1, \ldots, k\}$

$$c_i \equiv c_j (\mathrm{mod}\, d_{ij})$$
$$d_{ij} = \gcd(m_i, m_j)$$

that is

$$c_{k+1} = c_i (\mathrm{mod}\, d_{k+1,i})$$

We want to know if there an $x (\mathrm{mod}\, t')$ which is the solution for $x \equiv c (\mathrm{mod}\, t)$ and $x \equiv c_{k+1} (mod m_{k+1})$. That is whether the statement

$$c_{k+1} \equiv c [\mathrm{mod}\, \gcd(t, m_{k+1})]$$

is true or not.

## Relation between gcd and lcm operators

From chapter 22, exercise H4, let $a \star b = \gcd(a, b)$ and $a \circ b = \mathrm{lcm}(a, b)$ then it is trivial to show that

$$a \star (b \circ c) = (a \star b) \circ (a \star c)$$

and we know that the lcm operation is associative.

$$m_1 \circ m_2 \circ \cdots \circ m_k = m_1 \circ (m_2 \circ (\cdots \circ m_k))$$

so

$$\begin{aligned} m_{k+1} \star (m_1 \circ m_2 \circ \cdots \circ m_k) &= (m_{k+1} \star m_1) \circ (m_{k+1} \star (m_2 \circ \cdots \circ m_k)) \\ &= (m_{k+1} \star m_1) \circ (m_{k+1} \star m_2) \circ (m_{k+1} \star (m_3 \circ \cdots \circ m_k)) \\ &= (m_{k+1} \star m_1) \circ \cdots \circ (m_{k+1} \star m_k) \end{aligned}$$

That is

$$\gcd(\mathrm{lcm}(m_1, \ldots, m_k), m_{k+1}) = \mathrm{lcm}(\gcd(m_1, m_{k+1}), \ldots, \gcd(m_k, m_{k+1}))$$

## Proving equivalency holds under gcd for $k+1$

At the beginning it was stated that $\forall i \in \{1, \ldots, k\}$

$$c_{k+1} \equiv c_i (\mathrm{mod}\, d_{k+1,i})$$

and we also know that

$$c \equiv c_i (\mathrm{mod}\, m_i)$$
$$c - c_i = q m_i = q(s d_{k+1,i})$$
$$\implies c \equiv c_i (\mathrm{mod}\, d_{k+1,i})$$

## Generalizing lcm to Multiple Arguments

The lcm is defined as if $c = \mathrm{lcm}(a, b)$ then

1. $a \mid c$ and $b \mid c$
2. For any $x$ if $a \mid x$ and $b \mid x \implies c \mid x$

This can be generalized for any number of arguments in the lcm by noting that since $c = \mathrm{lcm}(x_1, x_2, \ldots, x_n)$ then $\forall i \in \{1, \ldots, n\}$ then 1. $x_i \mid c$ for 2., note that the common multiples of $\{x_1, \ldots, x_n\}$ form an ideal of $\mathbb{Z}$ by $\langle c \rangle = \langle x \rangle \cap \cdots \cap \langle x_n \rangle$, and so every common multiple is a multiple of $c$.

$\therefore$ any $v$ such that $\forall x_i \in X : x_i \mid v \implies c \mid v$.

## Solution $c \equiv c_i$ is also a Solution in the lcm of the gcds

From theorem 4, we generalize that

$$m_1 \mid x, \cdots, m_n \mid x \implies t \mid x$$

$$m_1 \mid (x-c), \cdots, m_n \mid (x-c) \implies t \mid (x-c)$$

$$x \equiv c(\mathrm{mod}\, m_1) \qquad \cdots \qquad x \equiv c(\mathrm{mod}\, m_n) \implies x \equiv c(\mathrm{mod}\, t)$$

where $t = \mathrm{lcm}(m_1, \ldots, m_n)$

Now note that

$$d_{k+1,1} \mid (c - c_i) \qquad \cdots \qquad d_{k+1,k} \mid (c - c_i) \implies \mathrm{lcm}(d_{k+1,1}, \ldots, d_{k+1,k}) \mid (c - c_i)$$

or

$$c \equiv c_i(\mathrm{mod}\, d_{k+1,1}) \qquad \cdots \qquad c \equiv c_i(\mathrm{mod}\, d_{k+1,k}) \implies c \equiv c_i[\mathrm{mod}\, \mathrm{lcm}(d_{k+1,1}, \ldots, d_{k+1,k})]$$

## There is a Common Solution for $c$ and $c_{k+1}$

So,

$$c \equiv c_i[\mathrm{mod}\, \mathrm{lcm}(\gcd(m_{k+1}, m_1), \ldots, \gcd(m_{k+1}, m_k))]$$

But we know that

$$\mathrm{lcm}(\gcd(m_{k+1}, m_1), \ldots, \gcd(m_{k+1}, m_k)) = \gcd(\mathrm{lcm}(m_1, \ldots, m_k), m_{k+1})$$

$$\implies c \equiv c_i[\mathrm{mod}\, \gcd(t, m_{k+1})]$$

where $t = \mathrm{lcm}(m_1, \ldots, m_k)$

And because of this, by theorem 3, because $\forall i \in \{1, \ldots, k\}, c \equiv c_i[\mathrm{mod}\, \gcd(t, m_{k+1})]$, there is an $x$ such that

$$x \equiv c(\mathrm{mod}\, t)$$

$$x \equiv c_{k+1}(\mathrm{mod}\, m_{k+1})$$

which because $x \equiv c(\mathrm{mod}\, t)$, this is also the solution for

$$x \equiv c_1(\mathrm{mod}\, m_1)$$

$$\ldots$$

$$x \equiv c_k(\mathrm{mod}\, m_k)$$

Furthermore this solution takes the form

$$\begin{aligned} x &\equiv c'[\mathrm{mod}\, \mathrm{lcm}(t, m_{k+1})] \\ &\equiv c'(\mathrm{mod}\, t') \end{aligned}$$

where $t' = \mathrm{lcm}(m_1, m_2, \ldots, m_k)$